Students Performance Prediction System

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Abstract

The Student Performance Prediction System is a machine learning-based predictive analytics application designed to estimate students' examination scores based on their academic and behavioral attributes. The project utilizes a supervised machine learning approach to identify relationships between study habits and academic performance.

Extra effort:

A web-based dashboard was also developed by me to allow users to input student information and receive real-time predictions.

Problem Statement

Educational institutions often struggle to identify students who may perform poorly before examinations. Early prediction of student performance enables teachers and students to take preventive actions. This project aims to predict examination scores using historical academic data.

Objectives

- Predict students' examination scores using Machine Learning.
- Perform data preprocessing and visualization.
- Evaluate machine learning performance.
- Build an interactive web dashboard. (extra effort)
- Deploy the prediction model using FastAPI through vercel. (extra effort)

Dataset

Source Link: Kaggle

Dataset Name: Student Performance Prediction Dataset **(10,000 Records)**

Features:

- Study Hours
- Attendance Percentage
- Sleep Hours
- Internet Usage
- Assignments Completed
- Previous Scores

Target:

- Exam Score

Technologies Used

- Python
- Google Colab (IDE)
- Scikit-learn
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Joblib
- FastAPI
- React.js
- Vercel (Deployment Frontend + Backend)

Machine Learning Algorithm

- Random Forest Regressor

Machine Learning Workflow

- Dataset Collection
- Data Cleaning

- Feature Selection
- Train-Test Split
- Model Training
- Model Evaluation
- Model Serialization

Evaluation Metrics

- **MAE (Mean Absolute Error)** : 5.84
- **MSE (Mean Squared Error)** : 65.42
- **RMSE (Root Mean Squared Error)** : 8.09
- **R^2 Score (R-Squared – coefficient of Determination)** : 0.72

Conclusion

The project successfully predicts students' examination scores using Machine Learning techniques. It demonstrates the complete machine learning lifecycle, including data preprocessing, model development, deployment, and frontend integration.

Links

- Dashboard
- Github (Python)
- Google Colab
- Kaggle Dataset
- Github (Dashboard)

AI Usage Declaration

AI Tools Used

- ChatGPT
- Claude AI

Purpose of AI Usage

AI tools were used as development assistants throughout the project.

Specifically:

- Understanding Machine Learning concepts.
- Used to find out tricks how to improve evaluation metrics.
- Assistance in debugging Python and React code.
- UI design suggestions.
- FastAPI integration guidance.
- Code optimization suggestions.

Sections Assisted by AI

- UI improvement suggestions
- To Show visuals using Matplotlib.
- To improvise the Evaluation Metrics.
- Debugging assistance
- Code explanation

Sections Implemented by Me

The following work was completed by me:

- Dataset selection from Kaggle
- Data preprocessing
- Exploratory Data Analysis (EDA)
- Machine Learning model training
- Model evaluation
- React frontend Development
- Fast-API backend development
- Frontend-backend integration
- Dashboard implementation
- Testing and deployment

Declaration

I declare that AI tools were used only as development assistants to improve productivity, debugging, documentation, and in improving the accuracy of model. All implementation, integration, testing, and final project decisions were completed by me. I understand and acknowledge the university's AI usage policy.